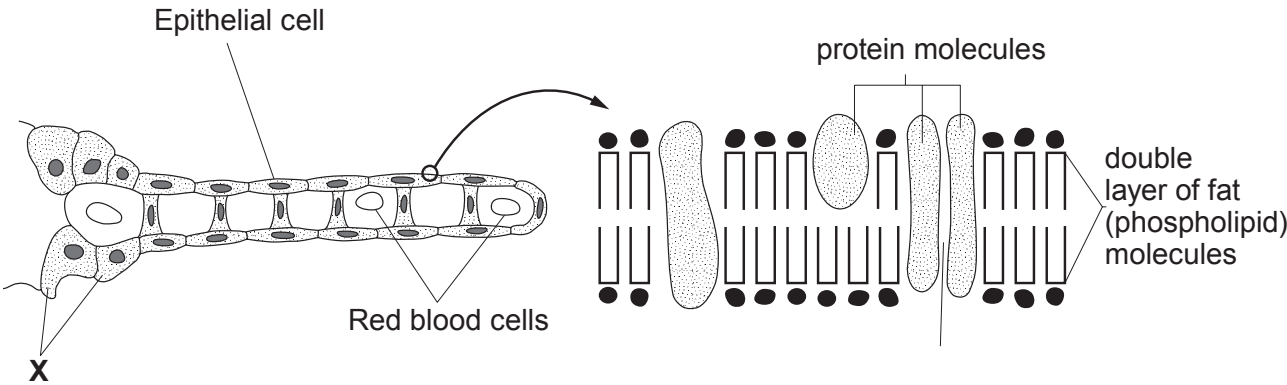


5. The primary role of the gills of fish is gas exchange. Oxygen, a non-polar molecule, passes from the surrounding water into the blood of the gills. The gills are also permeable to water and solutes, such as sodium ions and chloride ions.



- (a) Use the information provided to describe how oxygen and sodium ions cross the membrane. [4]

.....

.....

.....

.....

.....

.....



Examiner
only

(b) The data below shows the percentage ion composition of different aquatic habitats and of the blood plasma of two different fish.

		Percentage ion composition	
Fish	Habitat	Blood plasma of fish	Habitat
Flounder	Sea water	1.1	3.5
Carp	Fresh water	0.9	<0.1

Explain why it is important for the survival of the carp to produce large volumes of dilute urine. [4]

.....

.....

.....

.....

.....

.....

.....



- (c) (i) Specialised cells, labelled **X** on the diagram, are located in the epithelium of the gills. They are important in maintaining the ion composition of the blood plasma. Use the data to explain why this is necessary in the flounder. [3]

.....

.....

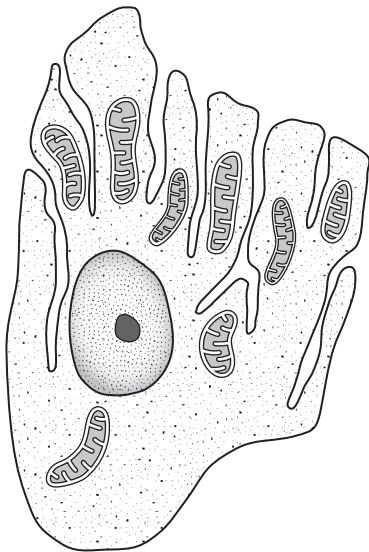
.....

.....

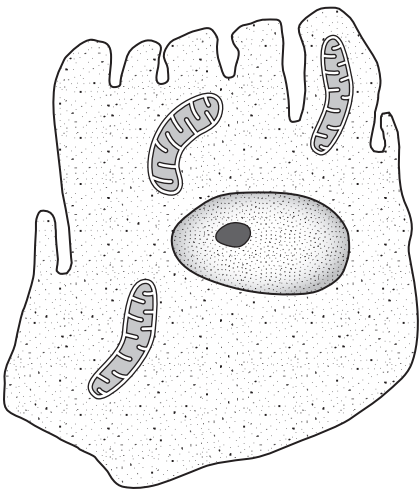
.....

- (ii) Salmon spend the majority of their lives in sea water. In order to reproduce salmon must migrate into fresh water rivers. During these migrations these specialised cells undergo structural changes.

The diagrams below are of these cells, taken from the salmon in the two different habitats. Giving reasons for your answer, suggest which cell was taken from the salmon in fresh water, and which was taken from the salmon whilst in sea water. [2]



Cell 1



Cell 2

.....

.....

.....

.....

.....



Examiner
only

(d) Acidification of fresh-water lakes, due to acid rain, has been linked to the death of fish such as carp. Scientists concluded that one of the causes of death in these animals is their inability to maintain blood plasma ion concentrations. Use the information to explain how they arrived at this conclusion. [4]

.....

.....

.....

.....

.....

.....

.....

17

